

Employment Passport

Every skill, wage, contract and review issued as a credential no single party can rewrite — and verifiable from anywhere in under 200 ms.

A broker arrives with a job in Saudi Arabia.

THE PROMISE

Alamgir Hossain had found nothing at home. The dalal spoke of a company, a salary, a future. There was no contract he could read, no licence he could check, no record anywhere that the promise had been made.

His family raised the fee the way most do — borrowing, and selling what they had. Bangladeshi workers pay BDT 3–5 lakh for a Saudi visa; the average across all corridors is 4.17 lakh, the highest in the world.

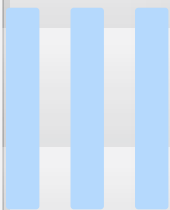


THE CAGE

**On arrival there is no job,
no employer, no iqama.**

**“The broker brought us and kept us
in a cage. After three months I
became illegal because I wasn’t
given an iqama. No work, no money
— I felt completely helpless.”**

— Alamgir Hossain, in Jeddah. Reported by
[Jago News](#), February 2026.



NO AUTHOR

Every promise made to him had no author.

“The agencies just bring us and abandon us,” he said. “No follow-up. No iqama. No job.”

Nobody had signed anything. There was no one to hold, nothing to point to, no record a stranger could check.

Alamgir did not need a database. He needed a promise that someone had to sign — and that no one could later erase. That is what we built.

1,130,757

workers left Bangladesh in 2025, returning USD 29.6 billion. Seventy-eight percent were classed less- or semi-skilled – and none carried portable, verifiable proof of what they can do.

The price of unverifiability

RECRUITMENT COST

4.17 lakh

BDT paid by the average migrant to get a job abroad — the highest worker-borne cost in the world.

PAYBACK

17.6 mo

of foreign earnings before that outlay breaks even.

ALREADY EXTRACTED

6–10k

BDT per worker to a destination-state verifier, mandatory in 33 sectors, on terms Bangladesh cannot renegotiate.

Workers will be verified either way. The question is who owns the record, who is paid for it, and whether the worker keeps a copy.

Six failure modes

01

Credential fraud

A forged welding certificate is cheap; checking a real one means phoning a training centre that may not answer.

02

Contract substitution

Sign one contract in Dhaka, receive a worse one — or none — on arrival. The original was never witnessed or timestamped.

03

Recruitment opacity

A rural jobseeker cannot tell a licensed agency from a fraudulent broker — so honesty earns the agency nothing.

04

Unverifiable reviews

An anonymous employer review is indistinguishable from a smear, so the market discounts it to zero.

05

Unenforceable labour law

Payroll and record both live in the accused employer's database, which the inspector must take on faith.

06

Career amnesia

A returning migrant's decade of experience is invisible at home; a garment worker loses tenure at every move.

“The problem is not that the data does not exist — it is that no party is trusted to hold it, and every party’s incentive is to manipulate it.”

W Worker — needs proof she can carry; cannot issue it herself.

A Agency — needs to be distinguishable from a fraud; no one believes its own claim.

E Foreign employer — needs to trust a stranger’s paper from a country it has no relationship with.

R Regulator — needs records it did not have to take on faith.

B Bank — needs a wage history to lend against; has only the employer’s word.

02

The solution

One identity, credential and anchoring layer. Five surfaces on top.

One ledger, five surfaces

SURFACE

QUESTION IT ANSWERS

BUILT FROM

Professional identity

What can this person provably do?

Skill, contract, wage credentials

Workplace transparency

What is it actually like to work here?

Verified-anonymous reviews

Corporate identity

Who is this company, and what does it owe?

Registry, ownership, wage disclosure

Operations Suite

How does a company run a compliant workforce?

The paid workflow layer

Labour-law enforcement

Does a violation reach the authority that can act?

Review signals routed to the ministry

Alamgir's journey, replayed

01	02	03	04	05	06
Skill certified	Broker checked	Application	Hired abroad	Wages anchored	Exit record
TTC issues a level credential; BMET clearance anchored.	He dials a USSD code: the agency's standing digest, before paying a taka.	Attested fields verified automatically; asserted fields signed by the agency.	Employer verifies the proof in seconds against the public anchor.	Every payslip co-signed by employer and bank.	Service record corroborates – or contradicts – what the agency asserted.
Nothing he was promised would have been unverifiable. On return, his wage history makes him bankable without a guarantor.					

Attested versus asserted

ATTESTED · POINTERS TO ANCHORED CREDENTIALS

Skill level · clearance · prior employers
· wage band · conduct findings

Verified automatically against the ledger. The agency cannot fake what it does not sign.

ASSERTED · THE AGENCY'S OWN SIGNED CLAIMS

Availability · undocumented experience
· languages · willingness to relocate

Unverifiable at screening — this is where the fraud lives. The agency stakes its standing on every word.

Every claim moved to the left column is risk the agency no longer carries: the issuance demand that solves cold start.

Verified anonymity

1 · COMMIT

Wallet → employer

Generates a per-employer link secret; sends only its commitment C. The employer never learns the secret.

2 · BIND

Employer → wallet

Issues an EmploymentProof with C signed inside as a blinded attribute — closing the fresh-secret attack.

3 · PROVE

Wallet → chaincode

Submits the review, a ZKP, and $n = \text{PRF}(\text{secret}, \text{employerDID})$. Reveals neither worker nor credential.

4 · ACCEPT ONCE

Chaincode

n unseen → one verified voice. n seen → rejected. Public digest: "12 verified reviewers · 74% recommend".

Why not a database

WHAT THE PLATFORM NEEDS

A record no office can fabricate

A history no one can rewrite

Proof nothing was withheld

Audit by a stranger, no account

A DATABASE

Admin inserts any row

Admin can UPDATE / DELETE

You see what you are shown

An API you must trust

THE LEDGER

Employer + bank co-sign, or it is not a record

Append-only, hash-chained

Merkle inclusion against a public root

Daily root on a public chain

Blockchain does not make dishonest people honest. It makes dishonesty permanent, attributable and expensive

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and protects the honest official.

03

Architecture

Hyperledger Fabric, permissioned. Nothing personal on-chain.

Six organisations, six channels

FOUNDING CONSORTIUM

Raft, one orderer each

MoEWOE / BMET — regulator
RJSC — corporate registry
BFIU — AML/CFT observer
BAIRA + agencies — industry
Partner banks — wage witness
Foundation — non-profit operator

CHANNELS

Visibility partitioned

anchor — hashes, DIDs, revocation
recruitment — orders, applications
payroll — wage anchors
corporate — registry, UBO proofs
reputation — reviews, standing
governance — founders only

INVARIANT

Nothing personal on-chain

On-chain: salted hashes, DIDs,
nullifiers, threshold proofs, status lists.
Off-chain: encrypted IPFS vault,
envelope-keyed to the worker's DID;
erasure by crypto-shredding. Daily 32-
byte Merkle root to a public chain, 4-
of-6 threshold-signed.

Endorsement, per schema

SCHEMA

WHO MUST SIGN

WHY

EmploymentContract

employer + agency

Neither party can invent a contract alone

WageEvent

employer + bank

Payer plus an independent financial witness

WorkplaceReview

worker ZKP + nullifier

Anonymity with a one-voice guarantee

MismatchAllegation

employer alone

Anchors as an allegation — never a finding

StandingAdjustment

regulator + observer, after 14 days

No party damages another's standing alone

A signer, never an owner

ROLE	CAN	CANNOT
Regulator (MoEWOE / BMET)	Co-endorse standing adjustments; issue clearances; read every channel	Write alone, delete, or alter any anchored record
Observer (BFIU)	Co-endorse with the regulator; audit flows	Initiate allegations or issue credentials
Issuers (TTCs, employers, agencies, banks)	Sign the facts they are responsible for	Sign for another party or revoke another's credential
Foundation (operator)	Run nodes, publish the daily root, maintain schemas	Hold a data key, sell reads, or change a quorum without 4-of-6



< 200 ms

to verify a Bangladeshi worker from Riyadh – no Fabric membership, no API key, no contact with any Bangladeshi institution. A cryptographic proof needs no jurisdiction.

04

The business

We sell writes and discovery. Never reads. The worker pays nothing, ever.

63

licensed agencies on the Operations Suite – 2.6% of the 2,400 – cover BDT 55 M of annual opex, at ~BDT 1,900 per placement.

Who pays, and for what

RECURRING · FUNDS OPERATIONS

Agencies, employers, banks

Operations Suite BDT 150–200 per managed worker per month (migration), BDT 25–40 domestic.

Bank adapter licence BDT 300–600k a year. Talent-search seats BDT 8–12k a month.

VOLUME-LINKED · SCALES WITH CORRIDOR

Deployments and clearances

BDT 1,000 per completed deployment workflow. BDT 50 per BMET clearance, bundled into the existing process.

Standalone issuance BDT 100 — free for Suite customers, so issuance is a reason to subscribe.

EPISODIC · ONCE NATIONAL

Corporate and policy events

Workforce transfers on merger, verified due diligence, buyer compliance packs: BDT 100k–1.5 M per event.

Ministry rule changes: BDT 40 per re-issued credential plus a BDT 2–5 M rollout package.

Unit economics

LINE	DERIVATION	PER AGENCY PER YEAR
Suite subscription	~200 managed workers × BDT 175/month × 12	BDT 420,000
Deployment logistics	~460 deployments × BDT 1,000	BDT 460,000
Total	recurring + volume-linked lines	~BDT 880,000
Cost per placement	BDT 880,000 ÷ 460	~BDT 1,900

Low single digits of an agency's margin, against a worker outlay of BDT 4.17 lakh. At BDT 55 M opex, 63 agencies break even.

Revenue trajectory

PHASE	PAYING ACCOUNTS	ANNUALISED REVENUE
1 · Corridor pilot	2 agencies, 1 TTC, 1 employer group	BDT 4–6 M
2 · Suite GA + regulator	65–80 agency and employer accounts	BDT 55–70 M · break-even
3 · Destination + corporate	200+ incl. manufacturers and banks	BDT 180–240 M
Mature, national scale	~BDT 47 bn of value created per year	BDT 500–700 M · under 2% captured

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Capturing under 2% is the point — our fee does not scale with the cost we remove.

Roadmap

MONTHS 0-6

Prototype

3 orgs, one corridor, one trade.

MONTHS 6-18

Corridor pilot

5,000 workers enrolled.

MONTHS 18-30

Suite GA + regulator

65-80 accounts. Break-even.

MONTHS 30-48

Destination + corporate

Verification exceeds issuance.

YEAR 4+

Regional

South Asian recognition.

Phases 0-1 cost BDT 65-80 M over eighteen months against ~BDT 5 M revenue — a gap of BDT 60-75 M, targeted at ILO/IOM, World Bank/IFC and bank co-investment. Fallback: launch the Suite commercially and fund the consortium from revenue.

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What would falsify this

- + The pilot's primary metric is employer-side verification calls per enrolled worker — not enrolment.
- + If that ratio stays near zero while credentials accumulate, the demand-side thesis has failed and we pivot to regulator-funded compliance infrastructure.
- + Second test: do workers like Alamgir actually consult agency standing before paying? If USSD lookups stay near zero, the mechanism's force must come from employers and the regulator.
- + An agency cannot pay for a higher standing, and no tier of the Suite confers one. The refusal is the product.

Team

01

Adib

Protocol & chaincode

02

SanjoyArchitecture &
cryptography

03

Samia

Research & policy

04

Alvee

Product & business model

Verify the proof.

Alamgir needed a promise with an author. A labour market that runs on provable facts instead of a broker's word – in which honesty becomes the cheapest available strategy.

Likely objections

"WHY NOT POSTGRESQL?"

A database has an owner. Here a wage event without the bank's co-signature is not an invalid record — it is not a record at all. Ordering, completeness and non-omission are provable to a stranger with no account.

"BROKERS ARE INFORMAL — HOW DO YOU REACH THEM?"

We don't regulate the dalal; we make his product worthless. A worker who can check a licensed agency's standing by USSD, and see which claims are attested, has no reason to pay for an unverifiable introduction.

"IF VERIFICATION IS FREE, WHERE IS THE MONEY?"

Writes and discovery: the Operations Suite, deployment workflows, clearances, talent search. 63 agencies break even. Free reads are what make the credential worth issuing.

"WHAT STOPS THE GOVERNMENT CAPTURING IT?"

The state is one signer of six. It cannot write alone, cannot delete, and its signatures are on a record nobody can move. The daily public root means even a captured consortium cannot rewrite history silently.